WTA Match Analysis

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Introduction:

Tennis has stood the test of time and has become one of the most popular sports in the world. By combining athletic skill, endurance, and strategy, each match is different and offers a complex interaction between the physical and environmental factors. Players train to take advantage of their physical attributes and skill to win on a variety of playing surfaces and match lengths.

We wanted to better understand the driving force behind match outcomes. Doing this for every tennis match ever, however would be too difficult and not relevant to the modern game. In order to minimize potential difficulties involving multiple players on the same team, we selected only women's singles matches. Additionally, we limited our data to matches in 2023 for up-to-date-ness and availability reasons. Thus, we posed the question: given surface type, player heights, player ages, and player handedness, which characteristic has the most significant impact on the probability that the better ranked player would win? The goal of this project is to determine which player or match characteristics—such as age difference, height difference, handedness, surface type, and tournament round—most strongly influence the probability that the higher-ranked player wins a women's professional tennis match. By defining an "expected winner" based on WTA ranking, we evaluate which factors, if any, help explain when expected outcomes occur and when upsets happen.

This project will analyze match-level data from the Women's Tennis Association (WTA) for the 2023 season. Each observation represents 1 match, which includes player attributes, set scores, and surface type. By modeling the relationship between our factors and match result, our study identifies which variables most strongly influence the winner of a tennis match.

Our motivation for this research stems from an interest in using sports analytics to understand better and predict performance outcomes. With events such as the 2025 Cincinnati Open ending August 2025, and becoming increasingly popular, predicting a winner can prove to be a fun and engaging exercise. Additionally, for sports bettors, being able to predict a winner based purely off of available data before a match could prove to be very profitable on sites such as Kalshi growing popularity.

Data Description:

The data we chose was the Women's Tennis Association data set from Awesomedata's GitHub repository (https://github.com/JeffSackmann/tennis_wta/blob/master/wta_matches_qual_itf_2023.csv) The data was created in 2023 and was collected from the International Tennis Federation. The data contains 34,323 observations and 49 variables. The variables of interest in our research include surface, winner and loser height, winner and loser age, and winner and loser hand. Each observation refers to a single match, not an individual player.

Table 1. Key Variables in the Tennis Dataset

Variable	Description	Type
`expwin`	1 if lower seeded player won, 0 otherwise (response variable)	Categorical
`minutes`	Duration of the match in minutes	Numeric
`surface`	Type of court surface (Clay, Hard, Grass, etc.)	Categorical
`winner_hand / loser_hand`	Handedness of the winner/loser (R, L, or U for unknown)	Categorical
`winner_age / loser_age`	Age of the winner/loser (in years)	Numeric
`winner_ht / loser_ht`	Height of the winner/loser (in centimeters)	Numeric
`winner_rank / loser_rank`	World ranking of the winner/loser at the time of the match	Numeric
`winner_rank_points / loser_rank_points`	Ranking points of the winner/loser	Numeric
`tourney_name`	Name of the tournament	Categorical
`tourney_level`	Tournament level (e.g., ITF, WTA 125, WTA 250)	Categorical
`tourney_date`	Starting date of the tournament	Date
`winner_seed / loser_seed`	Tournament seeding for the players (if applicable)	Numeric
`score`	Final match score (set-by-set)	Character
`round`	Stage of the tournament (Qualifying, stage of bracket format)	Numeric

Summary of the Data Most tournaments were played on hard or clay surfaces, with draw sizes typically of 32 players. The average winner age was 23.1 years (range: 14–51), while the average loser age was 22.9 years. Player heights were generally around 172 cm for winners and 171.8 cm for losers, though many values were missing. Right-handed players made up the majority of both winners and losers, while left-handed and unknown cases were less common.

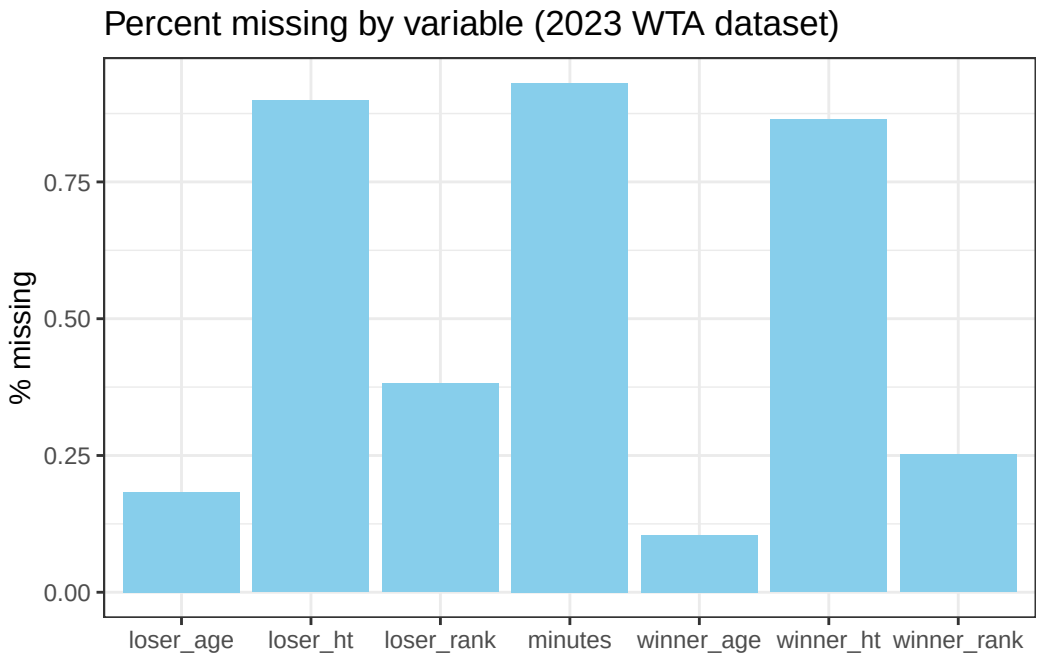
There are missing values in several variables, particularly for `winner_ht`, `loser_ht`, and match-level statistics such as aces and double faults.

During preprocessing, many variables, especially height and hand, contained substantial numbers of NAs. After restricting to the predictors required for our model and removing rows with missing values, the final analytic dataset consisted of 605 matches. This reduction was driven primarily by missing height and match time data rather than intentional filtering by tournament level or other match features.

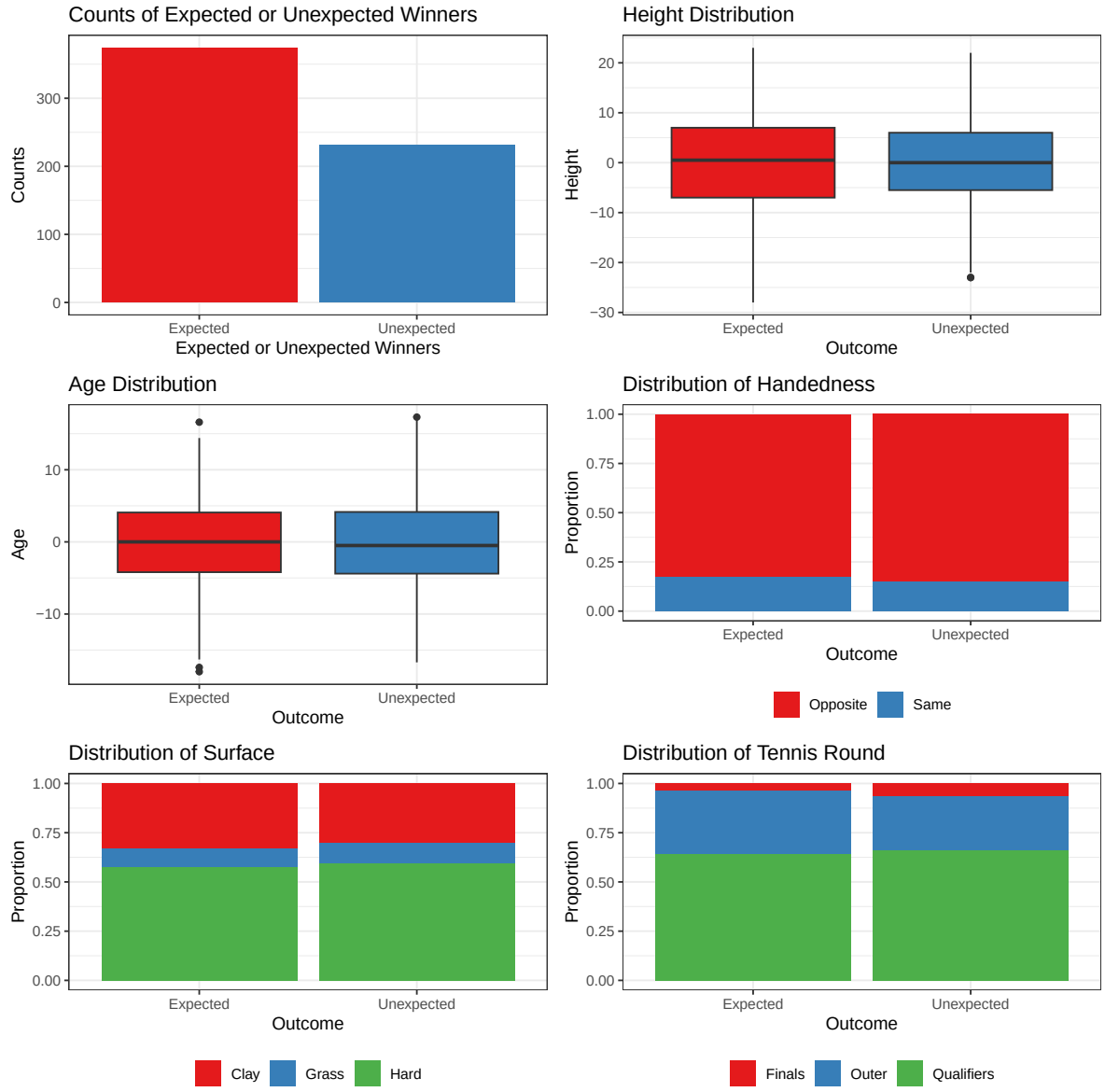
Each observation corresponds to a single match, not a single player, and includes the physical attributes of both players, ranking information, match conditions, and outcomes. To improve interpretability, we computed several difference variables (e.g., age_diff = winner_age – loser_age). These differences are numeric and continuous. The "winner_ / loser_" format in the raw data reflects the source structure, and the differences were computed directly from these values. Additionally, another conditional variable was created around handedness. If the handedness of the winner and loser were the same, then the variable `handed_diff` evaluated to 0. Finally, the `round` variables was re-leveled into "Qualifier" (for matches in the Qualification stage), "Outer" (for matches in the bracket stage before the semifinals), and "Finals" (for matches in the semifinals and finals). The first few rows are glimpsed below.

Then, the response variable was formulated: We defined expwin = 1 when the winner was expected to win (i.e., had a better (numerically lower) ranking), and expwin = 0 when the winner was an underdog.

expwin	height_diff	age_diff	handed_diff	surface	round
0	6	1.9	0	Clay	Finals
0	-2	5.1	0	Clay	Outer
0	9	-5.6	0	Clay	Outer
0	7	8.9	0	Clay	Outer
1	-6	-5.0	0	Clay	Outer
1	0	-4.4	0	Clay	Finals

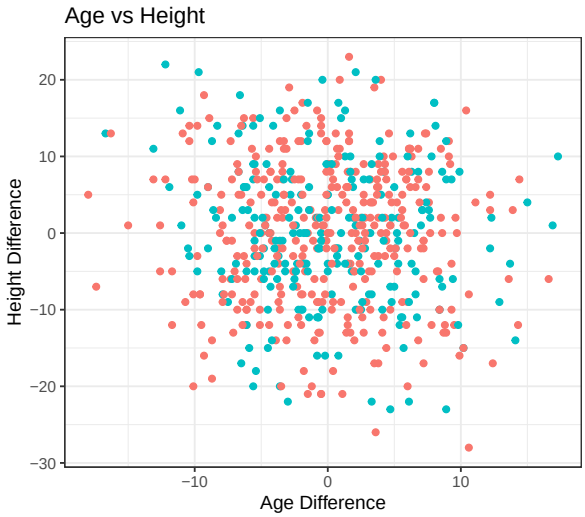

Percent missing by variable (2023 WTA dataset)
% missing
0.00
0.25
0.50
0.75
loser_age
loser_ht
loser_rank
minutes
winner_age
winner_ht
winner_rank

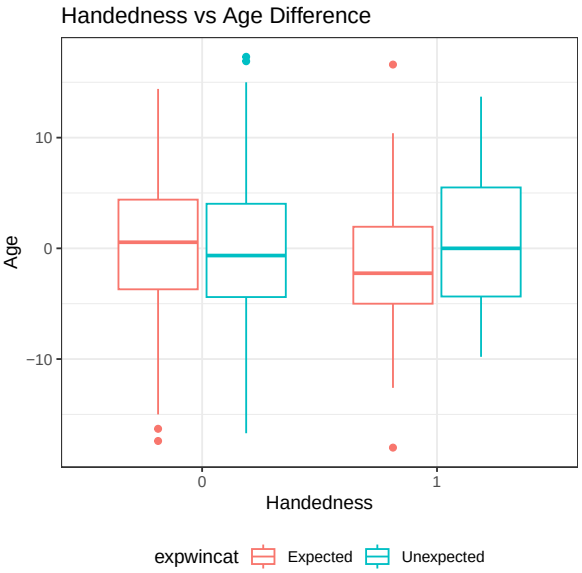
Univariate EDA:

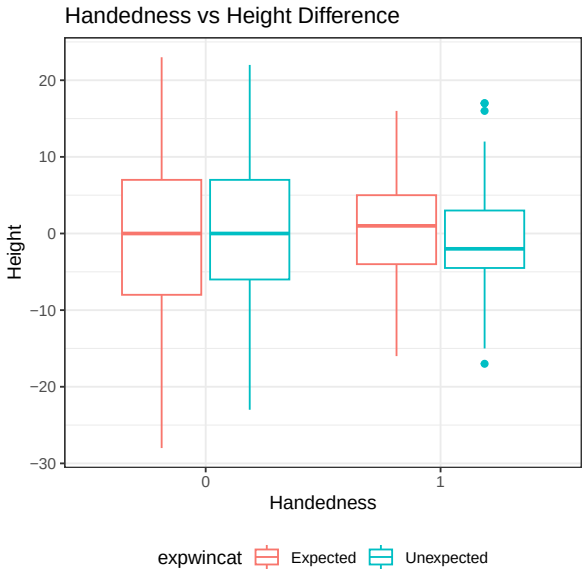


We observe that the dataset contains a larger number of expected winners compared to unexpected winners. This aligns with intuition, as better players tend to win more frequently in professional tennis. However, when we examine the distributions of the predictors relative to the response variable, no substantial differences emerge, which is also observed with no meaningful differences between handedness, surface, and tournament round. The boxplots for age and height differences show overlapping distributions for expected and unexpected winners, suggesting that neither variable appears to systematically favor one outcome over the other.

Bivariate EDA:







Examining the bivariate relationships among our predictors, we observe largely similar patterns across variables. Age differences and height differences appear randomly distributed, with no clear directional trends or clustering. The distributions of handedness relative to both age and height differences also show no meaningful separation, suggesting minimal association among these characteristics. Overall, the variables seem largely independent of one another. While there is a slight indication that rank may increase with age, this relationship is modest and will be further evaluated using VIF diagnostics later in the analysis.

Methodology

Following the EDA, it remained unclear which predictors should constitute an "optimal" model, as all appeared theoretically relevant. To address this, we implemented a workflow that fit all possible models excluding interactions; interactions were omitted due to computational constraints, as including all second-order interactions among six predictors would require 63 additional models. Additionally, the bivariate EDA demonstrated a lack of collinearity between models or interactions.

For each model, we computed AIC and AUC, standardized both metrics using $\frac{(max(AIC)-AIC)}{(max(AIC)-min(AIC)}$ and $\frac{(AUC-min(AUC))}{(max(AIC)-min(AIC)}$ and combined them with equal weighting to give a balanced selection criterion. This procedure initially identified `expwin` ~ `round` as the best model. This was originally selected as not all models could be directly compared utilizing drop-in-deviance testing, as many were not subsets of one another.

However, the AIC values varied by only 9.156463 across all models — an insignificantly small range given overall values near 800 — indicating that AIC offered little discriminatory power. Consequently, we prioritized AUC alone, which yielded a more informative model: `expwin` ~ `age_diff` + `handed_diff` + `round`.

Results and Conclusion

Coefficients:

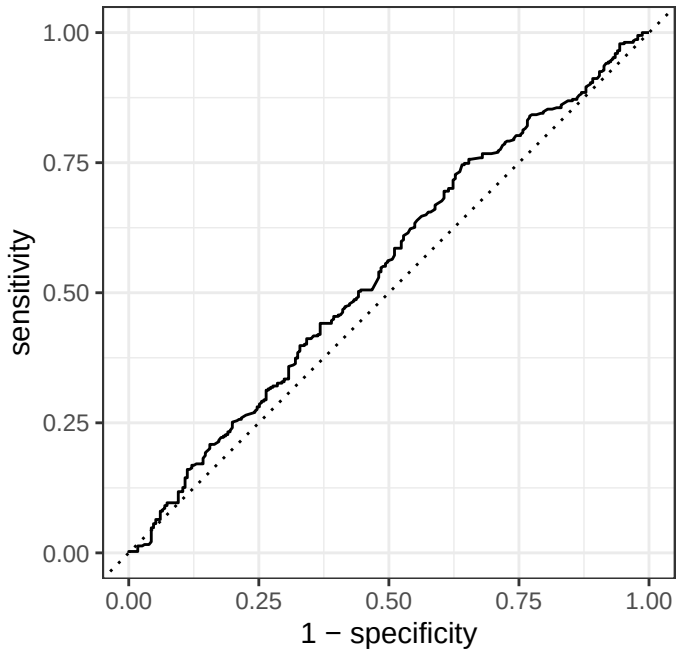
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.49532	0.08914	5.55689	0.00000	0.32062	0.67003
age_diff	0.00010	0.00334	0.02886	0.97698	-0.00645	0.00664
handed_diff1	0.03474	0.05363	0.64776	0.51739	-0.07037	0.13985
roundOuter	0.15358	0.09589	1.60152	0.10979	-0.03437	0.34152
roundQualifiers	0.10926	0.09219	1.18515	0.23643	-0.07143	0.28996

Optimal Model Evaluation

AIC	AUC
811.3485	0.5453156

VIF

names	x
age_diff	1.004
handed_diff1	1.006
roundOuter	4.943
roundQualifiers	4.945



The ROC curve for the selected logistic regression model shows performance only slightly above random guessing. The area under the curve (AUC = 0.54) indicates that the model has very weak discriminatory ability, meaning it is only marginally better than a coin flip at distinguishing expected winners from unexpected winners. The ROC curve itself remains close to the diagonal reference line, which represents a classifier with no predictive power. This suggests that the included predictors, age difference, handedness difference, and round, do not provide meaningful signal for predicting whether a match's winner is expected or unexpected. In other words, the model struggles to rank players by their probability of winning based on these features alone. This result is consistent with earlier exploratory analyses, which showed substantial overlap in predictor distributions across outcome categories. Overall, the low AUC indicates substantial limitations in the model's usefulness for prediction in this context.

Our final model yielded an AIC of 811.35 and an AUC of 0.545, indicating predictive performance only marginally better than chance. The AIC was similar to those of alternative model specifications, and all VIF values were below 5, suggesting that multicollinearity among predictors is not a concern. While we treated each match as independent and identically distributed, we acknowledge that some players faced each other multiple times; removing repeated matchups would have introduced additional selection issues, so all matches were retained.

Interpreting the model using exponentiated coefficients, we find that the effects of the included predictors are negligible in practical terms. Specifically, for each year increase in age difference, the expected winner's odds of winning increased by $e^{0.0001} = 1.0001$ (95% CI 0.9936- 1.0067) times. Differences in handedness multiply the odds by $e^{0.03474} = 1.0353$ (95% CI 0.932-1.150). Early-round matches (roundOuter) multiply the odds by $e^{0.1535} = 1.166$ (95% CI 0.966-1.407),

and qualifying-round matches (roundQualify) multiply the odds by $e^{0.10926} = 1.115$ (95% CI 0.931-1.336). However, none of these effects reach conventional statistical significance (alpha = 0.05), indicating that we cannot conclude these factors meaningfully influence the probability of an expected winner prevailing.

The univariate EDA revealed heavy overlap in age and height differences between expected and unexpected winners. These patterns mirror the weak effects observed later in the logistic regression. This makes sense: if the distributions of predictors are nearly identical, the predictors cannot produce meaningful separation of the two outcome groups. This early visual evidence helps contextualize why the final model achieves only minimal predictive power.

We also elected not to pursue any interaction effects, as we found a majority of our bivariate analysis did not demonstrate any form of interaction or collinearity. Future efforts may be focused here, along with utilizing other match-level data to create a stronger model.

Overall, the model provides limited predictive power, with an AUC only slightly above 0.5. This suggests that static player attributes such as age, height, handedness, and surface type contribute very little to determining match outcomes. Instead, performance likely depends more on dynamic factors that are not captured in the dataset, such as player skill, strategy, fitness, in-match decision-making, and psychological factors. These results highlight the complexity of professional tennis, where observable demographic or physical characteristics alone are insufficient to reliably predict match outcomes.

Citation:

Sackmann, Jeff. 19 March 2024. WTA Tennis Rankings, Results, and Stats. creative commons. https://github.com/JeffSackmann/tennis_wta/blob/master/README.md